

Lucas Fernandez Sarmiento

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Professional Summary

Machine-learning researcher and PhD candidate in theoretical physics at Carnegie Mellon, using statistical mechanics, probabilistic modeling, and large-scale experiments to study high-dimensional learning systems. Expected PhD completion: August 2027; available to start September 2027. Author of a single-author ICML 2026 paper on neural-network training dynamics, dropout scaling laws, and compute-efficient scheduling, validated across MLPs, ResNets, and Vision Transformers. Experience developing robust LLM aggregation, sequence-modeling, optimization, and decision-system pipelines using PyTorch, Hugging Face, scikit-learn, and NumPy, with broader interests in scalable generative and recommendation systems.

Education

Carnegie Mellon University PhD Candidate, Theoretical Physics (GPA: 4.00/4.00) <i>Expected August 2027</i>	<i>2022–present</i>
Imperial College London MSc, Theoretical Physics (Distinction)	<i>2021–2022</i>
Imperial College London BSc, Theoretical Physics (First Class, Dean’s List)	<i>2018–2021</i>

Research and Professional Experience

PhD Researcher <i>Carnegie Mellon University</i> <i>Advisors: Riccardo Penco and Rachel A. Rosen</i>	<i>Pittsburgh, PA</i> <i>Aug 2022–present</i>
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- Derived a mean-field theory of dropout in stochastic neural-network dynamics, predicting scaling laws and depth-dependent schedules for compute-efficient training. Across MLP, ResNet, and ViT models on CIFAR-10/100, improved accuracy, lowered test loss by 18–35%, and reduced compute by up to 65% at matched accuracy.
- Developed and evaluated robust aggregation methods for adversarial multi-agent LLM debate under asymmetric information, comparing majority voting, linear and logarithmic pooling, and Hedge; ensembles of open-weight models outperformed Claude Sonnet 4.6 on single-answer question-answering tasks at approximately 70% lower inference cost.
- Built reproducible Python workflows for large-scale ML and simulation experiments, including PyTorch training loops, statistical diagnostics, scaling analysis, hyperparameter sweeps, and HPC/cluster execution.
- Extended mean-field and signal-propagation methods to residual networks, studying depth-dependent residual scaling and improved He initialization for stable training at large depth.
- Built Python workflows for large-scale tensor-network simulations, running DMRG experiments with scaling analysis and convergence diagnostics.

Adjunct Lecturer <i>Universidad de las Hespérides</i>	<i>Madrid, Spain</i> <i>Apr 2025–Dec 2026</i>
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- Designed and taught applied mathematics modules covering linear algebra, calculus, differential equations, and constrained optimization, with applications to portfolio-risk optimization.
- Developed Spanish ML lecture notes on data matrices, probability, PCA, SVD, least squares, Ridge/Lasso, logistic regression, cross-validation, and backpropagation for business and optimization examples.

Co-founder and Technical Lead <i>SteamAI</i>	<i>Luxembourg</i> <i>Jun 2024–Aug 2024</i>
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- Built multivariate time-series forecasting and constrained-control models for refinery steam networks under stochastic demand, combining autoregressive demand modelling with pressure-constrained optimization to improve operational efficiency.

Applied ML and GenAI Projects

- Forecasting and ranking simulation: Built an end-to-end probabilistic forecasting pipeline using Understat xG/match data, pandas/NumPy feature engineering, scikit-learn score models, and Monte Carlo simulations to evaluate calibrated outcome probabilities and ranking quality.
- QLoRA project: Implemented GRU, LSTM, and Transformer translation models and parameter-efficient Llama 2 7B fine-tuning with QLoRA in PyTorch.

Technical Skills

Deep Learning and Language Models: Transformers, attention, ViTs, ResNets, LSTMs, CNNs, LoRA and QLoRA, PyTorch, Hugging Face.

Probabilistic Modeling and Decision Systems: Monte Carlo simulation, calibration, stochastic processes, robust aggregation, no-regret learning, game theory, MDPs, multi-agent learning.

Optimization: Convex and constrained optimization, projected-gradient methods, cost-aware inference.

Programming and Computing: Python, NumPy, pandas, scikit-learn, Git, AWS, asyncio, SLURM/HPC, Mathematica.

Coursework: CMU ML (10-701), CMU AI Agents (11-768), CMU Adv. ML & Game Theory (17-759).

Selected Publications and Manuscripts

- Lucas Fernández-Sarmiento. *Dropout Universality: Scaling Laws and Optimal Scheduling at the Edge-of-Chaos* [🔗](#). Single-author paper accepted to ICML 2026; develops a predictive theory of dropout, regime changes in neural-network dynamics, and compute-efficient generalization through optimal scheduling.
- Keltin Grimes, Lucas Fernández-Sarmiento. *Don't Debate, Interrogate! Robust AI Debate Despite Deceptive Agents* (under preparation). Studies robust aggregation for multi-agent LLM systems under correlated, non-IID judge errors.
- Lucas Fernández-Sarmiento, Sophie Merchant, Rupert Mitchell. *Mean-Field Theory for ResNets: Depth-Dependent Residual Connections and Improved He Initialization* (under preparation).
- Lucas Fernández-Sarmiento, Riccardo Penco, Rachel A. Rosen. *On Entropy Bounds for Irrelevant Operators*. Accepted to JHEP. [arXiv:2508.14978](#) [🔗](#).
- Lucas Fernández Sarmiento, Irvin Martínez Rodríguez. *Broken Weyl Gravity: Dynamical Torsion and Generalized Proca*. [arXiv:2510.08778](#) [🔗](#).

Conference Presentations

- Scheduled to attend and present at *ICML 2026*.
- Expected presenter, *IAIFI Summer Workshop 2026* (upcoming; abstract submitted): *Dropout Universality: Scaling Laws and Optimal Scheduling at the Edge-of-Chaos*.
- Presented physics research at COSMO 2025 and the Cargèse BSM conference/summer school; attended *When QFT Meets QI*, *Contours 2026*, and *Amplitudes London 2026*.

Awards & Honors

2026–2027	CMU Presidential Fellowship, Mellon College of Science, funded by Tata Consultancy Services
2026	NSF ACCESS Compute Allocation, 200,000 ACCESS credits (~2,000 H100 GPU-hours; PI)
2025	CMU Professional Engagement Award
2023	Rafael del Pino Foundation Excellence Scholarship (\$28,600)
2021	UROP Bursary Prize, Imperial College London (~\$3,000)
2019, 2020	Dean's List, Imperial College London
2018	Finalist, Big Bang Young Scientist of the Year (UK)
2017	Gold CREST Award (Jaguar Land Rover research)